Learning to Walk Randomly

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Abstract

This paper proposes to learn to walk by randomly sampling actions. This abstract should be short and concise, about 8-10 lines long.

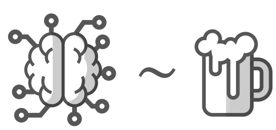
1 Methodology

This paper proposes training a bipedal robot to learn to walk by sampling actions a from a stochastic policy $a \sim \pi(s, \theta)$. The policy proposed is to iteratively sample our four actions from a continuous uniform distribution. These correspond to the bipedal walker agent action space (referenced here):

	Name	**Min**	**Max**
0	Hip 1 (Torque/velocity)	-1	$+1$
1	Knee 1 (Torque/velocity)	-1	$+1$
2	Hip 2 (Torque/velocity)	-1	$+1$
3	Knee 2 (Torque/velocity)	-1	$+1$

Table 1: The continuous action space is between -1 and $+1$ so `torch.tanh` is sensible in the last actor layer. You can make tables like this using this generator .

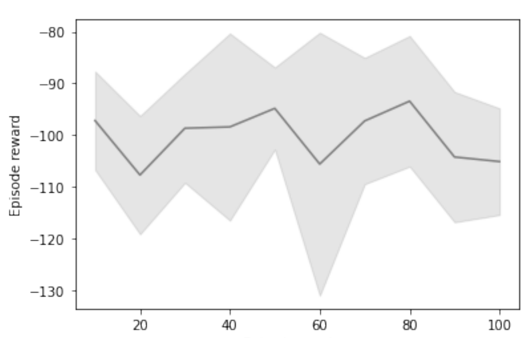
The methodology should be very concise and the mathematical notation should try to follow the ICLR conference guidlines . If you are not familiar with LaTeX, you can use this online LaTeX equation editor . You may want to include an architectural diagram:



The diagram above was created using Inkscape and exported to a PDF. This was then uploaded to the figures directory on the left.

2 Convergence Results

The model does not converge or get any positive scores in our experiments, although there is a non-zero probability that it will eventually find the optimal policy. However, this is not sample efficient and is too slow to be considered practical.



3 Limitations

After several hours the agent is still unable to walk.

Future Work

In the future, it would be good to run for more than 100 episodes and to try and train a more modern method such as TD3 [1].

References

[1] Scott Fujimoto, Herke Hoof, and David Meger. "Addressing function approximation error in actor-critic methods". In: *International Conference on Machine Learning*. PMLR. 2018, pp. 1587–1596.